Number Sense and Operations

MA.912.NSO.1 *Generate equivalent expressions and perform operations with expressions involving exponents, radicals or logarithms.*

MA.912.NSO.1.1

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Benchmark Clarifications:
Clarification 1: Instruction includes the use of technology when appropriate.
Clarification 2: Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.
Clarification 3: Instruction includes converting between expressions involving rational exponents and expressions involving radicals.
Clarification 4: Within the Mathematics for Data and Financial Literacy course, it is not the expectation to generate equivalent numerical expressions.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.5.3
- MA.912.FL.3.2

Terms from the K-12 Glossary

- Base
- Exponent
- Expression

Vertical Alignment

Previous Benchmarks

- MA.8.NSO.1.3

Next Benchmarks

- MA.912.NSO.1.6

Purpose and Instructional Strategies

In grade 8, students generated equivalent numerical expressions and evaluated expressions using the Laws of Exponents with integer exponents. In Algebra I, students work with rational-number exponents. In later courses, students extend the Laws of Exponents to properties of logarithms.

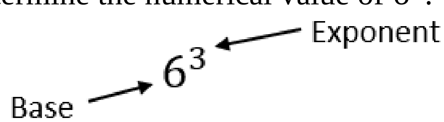
- Instruction includes using the terms Laws of Exponents and properties of exponents interchangeably.
- Instruction includes student discovery of the patterns and the connection to mathematical operations and the inverse relationship between powers and radicals *(MTR.5.1)*.
- Problem types include having a fraction, integer or whole number as an exponent.
- Students should make the connection of the root being equivalent to a unit fraction exponent *(MTR.4.1)*.
  - For example, $\sqrt[3]{8} = \sqrt[3]{2^3}$ is equivalent to the equation $\sqrt[3]{8} = (2^3)^{\frac{1}{3}}$ which is equivalent to the equation $\sqrt[3]{8} = 2^{\frac{3}{1}\cdot\frac{1}{3}}$ which is equivalent to the equation $\sqrt[3]{8} = 2^1$ which is equivalent to the equation $\sqrt[3]{8} = 2$.
- When evaluating, students should be encouraged to approach from different entry points and discuss how they are different but equivalent strategies *(MTR.2.1)*.
  - For example, if evaluating $(-27)^{\frac{2}{3}}$ students can either take the cube root of -27 first or raise -27 to the second power first.

Common Misconceptions or Errors

- Students may not understand the difference between an expression and an equation.
- Students may try to perform operations on bases as well as exponents.
- Students may multiply the base by the exponent instead of understanding that the exponent is the number of times the base occurs as a factor.
- Students may not truly understand exponents that are zero or negative.

Strategies to Support Tiered Instruction

- Teacher provides a review of the relationship between the base and the exponent by modeling an example of operations using a base and exponent.
  - For example, determine the numerical value of 6^3.



6^3 which is equivalent to $6 \cdot 6 \cdot 6$ which is equivalent to 216.

- Teacher provides exploration of the rules of exponents through patterns. A strategy for developing meaning for integer exponents by making use of patterns is shown below:

Patterns in Exponents	
5^5	$5 \cdot 5 \cdot 5 \cdot 5 \cdot 5$
5^4	$5 \cdot 5 \cdot 5 \cdot 5$
5^3	$5 \cdot 5 \cdot 5$
5^2	$5 \cdot 5$
5^1	5
5^0	1
5^{-1}	$\frac{1}{5}$
5^{-2}	$\frac{1}{5 \cdot 5}$
5^{-3}	$\frac{1}{5 \cdot 5 \cdot 5}$
5^{-4}	$\frac{1}{5 \cdot 5 \cdot 5 \cdot 5}$
5^{-5}	$\frac{1}{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}$

- Teacher provides exploration of the rules of rational exponents through patterns. A strategy for developing meaning for rational exponents by making use of patterns is shown below:

Patterns in Rational Exponents		
$5^{\frac{5}{2}}$	$\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}$	$\sqrt{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}$
$5^{\frac{4}{2}}$	$\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}$	$\sqrt{5 \cdot 5 \cdot 5 \cdot 5}$
$5^{\frac{3}{2}}$	$\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}$	$\sqrt{5 \cdot 5 \cdot 5}$
$5^{\frac{2}{2}}$	$\sqrt{5} \cdot \sqrt{5}$	$\sqrt{5 \cdot 5}$
$5^{\frac{1}{2}}$	$\sqrt{5}$	$\sqrt{5}$
$5^{\frac{0}{2}}$	$\sqrt{1}$ or 1	$\sqrt{1}$ or 1
$5^{-\frac{1}{2}}$	$\frac{1}{\sqrt{5}}$	$\sqrt{\frac{1}{5}}$ or $\frac{1}{\sqrt{5}}$
$5^{-\frac{2}{2}}$	$\frac{1}{\sqrt{5} \cdot \sqrt{5}}$	$\sqrt{\frac{1}{5 \cdot 5}}$ or $\frac{1}{\sqrt{5 \cdot 5}}$
$5^{-\frac{3}{2}}$	$\frac{1}{\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}}$	$\sqrt{\frac{1}{5 \cdot 5 \cdot 5}}$ or $\frac{1}{\sqrt{5 \cdot 5 \cdot 5}}$
$5^{-\frac{4}{2}}$	$\frac{1}{\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}}$	$\sqrt{\frac{1}{5 \cdot 5 \cdot 5 \cdot 5}}$ or $\frac{1}{\sqrt{5 \cdot 5 \cdot 5 \cdot 5}}$
$5^{-\frac{5}{2}}$	$\frac{1}{\sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5} \cdot \sqrt{5}}$	$\sqrt{\frac{1}{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}}$ or $\frac{1}{\sqrt{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}}$

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.5.1)

Part A. Think about when solving an equation with a radical. What is the inverse operation of a square root? Of a cube root?

Part B. Given the expression $\sqrt[3]{27}$, express 27 as a prime number with natural-number exponent.

Part C. How can we use the information from Part A and B to convert $\sqrt[3]{27}$ to exponential form?

Instructional Task 2 (MTR3.1, MTR.4.1)

Part A. Evaluate $64^{\frac{1}{3}}$ by first writing 64 as a power of 2 and using the properties of exponents.

Part B. Evaluate $64^{\frac{1}{3}}$ using a calculator.

Part C. Explain your process in both Part A and Part B. Define powers with fractional exponents in your own words.

Instructional Task 3 (MTR.3.1, MTR.5.1)

Part A. Given $f(x) = 32^x$, evaluate $f(0)$, $f(0.2)$, $f(0.4)$, $f(0.8)$ and $f(1)$ without the use of a calculator.

Part B. Graph the function f in the domain $0 \leq x \leq 1$.

Part C. Between which two values in Part A would $f(0.5)$ be? Which one would it be closer to on the graph and why?

Instructional Items

Instructional Item 1

Evaluate the numerical expression $(64)^{\frac{4}{3}}$.

Instructional Item 2

Rewrite $8^{0.5} \cdot 2^{\frac{2}{5}}$ as a single power of 2.

Instructional Item 3

Choose all of the expressions that are equivalent to $7^{\frac{5}{12}}$

a. $\left(49^{\frac{1}{3}}\right)\left(7^{-\frac{1}{4}}\right)$

b. $\left(7^{\frac{2}{3}}\right)\left(7^{-\frac{1}{4}}\right)$

c. $7\left(7^{-\frac{1}{4}}\right)$

d. $\sqrt[5]{7^{12}}$

e. $\sqrt[12]{7^5}$

Instructional Item 4

Evaluate the numerical expression $\left(-\frac{729}{64}\right)^{-\frac{2}{3}}$.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*